

Tricyclic Antidepressant Overdose



Section I: Scenario Demographics

Scenario Title:	Tricyclic Antidepressant (TCA) Overdose		
Date of Development:	26/08/2014		
Target Learning Group:	☐ Juniors (PGY 1 - 2)	☒ Seniors (PGY $\geq$ 3)	☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Martin Kuuskne
Affiliations/Institution(s):	McGill University
Contact E-mail (optional):	martin.kuuskne@mail.mcgill.ca

Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To recognize and treat and patient presenting after an intentional TCA overdose.
CRM Objectives:	1) Allocate attention wisely between vital signs, history, physical exam, EKG and treatment response in the setting of TCA toxicity. 2) Prevent and manage fixation error in the setting of a patient with altered mental status and fever by maintaining and addressing a wide differential diagnosis.
Medical Objectives:	1) Recognize and appropriately manage a patient in pulseless electrical activity (PEA) arrest. 2) Apply an approach, either rhythm-based or the traditional Hs & Ts, to aid in the management of the underlying cause of PEA. 3) Administer sodium bicarbonate therapy with appropriate treatment targets in the treatment of TCA toxicity with ECG findings of sodium channel blockade.

Case Summary: Brief Summary of Case Progression and Major Events

A 27-year-old male presents to the emergency department with altered mental status after an intentional Amitriptyline overdose. He is found to have a wide QRS complex and an anticholinergic toxidrome. The patient deteriorates into PEA arrest necessitating standard ACLS and intravenous sodium bicarbonate therapy.

References

- Nelson, L., Lewin, N., Howland, M., & Et al. (2011). Chapter 73 - Cyclic Antidepressants. In *Goldfrank's Toxicologic Emergencies* (9th ed.). New York: McGraw-Hill Medical.
- Littmann, L., Bustin, D., Haley, M. A Simplified and Structured Teaching Tool for the Evaluation and Management of Pulseless Electrical Activity, *Med Princ Pract* 2014;23:1-6, DOI: 10.1159/000354195
- Nikson, C. (2009, October 30). Tricyclic antidepressant (TCA) toxicity. Retrieved August 28, 2014, from <http://lifeinthefastlane.com/toxicology-conundrum-022/>



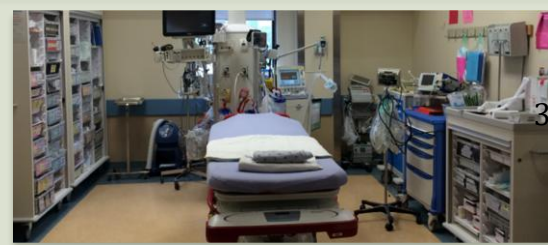
Tricyclic Antidepressant Overdose



Section IV: Scenario Script

A. Clinical Vignette: To Read Aloud at Beginning of Case			
You are working an evening shift in a tertiary care hospital emergency department. A 27-year-old male is brought to the resuscitation bay with a decreased mental status.			
B. Scenario Cast & Realism			
Patient:	☐ Computerized Mannequin ☒ Mannequin ☐ Standardized Patient ☐ Hybrid ☐ Task Trainer	Realism: <i>Select most important dimension(s)</i>	☒ Conceptual ☐ Physical ☐ Emotional/Experiential ☐ Other: ☐ N/A
Confederates		Brief Description of Role	
Mother	Hovers around bedside repetitively asking "Is David alright?" Gives HPI and review of systems if asked. If asked about medications in the house, states that she was recently started on amitriptyline for chronic back pain.		
C. Required Monitors			
☒ EKG Leads/Wires	☒ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☒ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
D. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☒ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☒ Other: NG tube	
☐ Intraosseous Set-up	☐ LMA	☒ Other: Activated Charcoal	
E. Moulage			
Mother confederate has a bag of her home medications that includes an empty amitriptyline pill container.			
F. Approximate Timing			
Set-Up:	5 min	Scenario:	12 min
		Debriefing:	15 min

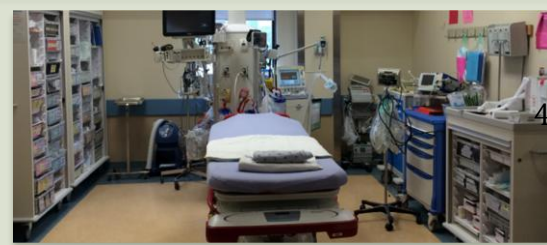
Tricyclic Antidepressant Overdose



Section V: Patient Data and Baseline State

A. Patient Profile and History					
Patient Name: David McGibbons		Age: 27		Weight: 70kg	
Gender: ☒ M ☐ F		Code Status: Full			
Chief Complaint: Altered Mental Status					
History of Presenting Illness: Found by his mother in his room having what looked like a seizure for 2 minutes. He was not responding and somnolent. He was last seen acting normally 2 hours prior to the event.					
Past Medical History:		None		Medications: None	
Allergies: NKDA					
Social History: Recreational EtOH and Marijuana use.					
Family History: Non-contributory					
Review of Systems:		CNS:		Normal	
		HEENT:		Normal	
		CVS:		Normal	
		RESP:		Normal	
		GI:		Normal	
		GU:		Normal	
MSK:		Normal		INT: Normal	
B. Baseline Simulator State and Physical Exam					
☐ No Monitor Display		☒ Monitor On, no data displayed		☐ Monitor on Standard Display	
HR: 122/min		BP: 109/71		RR: 14/min O ₂ SAT: 92%	
T: 38.1°C		Glucose: 6.5 mmol/L		GCS: 7 E1-noneV2-sounds M4-withdraws	
General Status: Depressed mental status.					
CNS:		Pupils 5mm bilat, poorly reactive. Non-focal exam of face/UE/LEs , 3+ reflexes, symmetrical.			
HEENT:		Normal			
CVS:		Tachycardia, no extra heart sounds			
RESP:		GAEB, no adventitia			
ABDO:		Soft, absent bowel sounds			
GU:		Palpable bladder			
MSK:		Normal		SKIN: Flushed skin	

Tricyclic Antidepressant Overdose



Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: Sinus rhythm + Wide QRS HR: 122/min BP: 109/71 RR: 14/min O ₂ SAT: 92 % T: 38.1°C	Altered mental status, minimally responsive, GCS: 7 (E1V2M4)	Learner Actions ☐ Monitors/Full vitals ☐ Hx/PE ☐ IV access ☐ Supplemental O ₂ ☐ EKG #1 ☐ IV NS Bolus ☐ Lab investigations ☐ IV NaHCO ₃ Bolus ☐ Induction medication ☐ Paralytic medication ☐ Intubation	Modifiers - Supplemental O ₂ → ΔO ₂ SAT to 97% - IV NaHCO ₃ bolus → change waveform from wide to narrow QRS complex. Change waveform back to wide QRS after 1 min Triggers - Intubation → 2. Intubation - 3 minutes → 3. PEA
2. Intubation HR → 130 BP → 90/65 O ₂ SAT → 97% <u>Display</u> (if possible) V _t : 500ml RR: Set FiO ₂ : 1.0 EtCO ₂ = 50, normal waveform	Easy intubation.	Learner Actions ☐ IV NS Bolus ☐ IV Vasopressors ☐ IV antibiotics (CNS coverage) ☐ OG/NG tube ☐ Activated Charcoal ☐ IV NaHCO ₃ Bolus ☐ EKG #2 ☐ Post intubation CXR	Modifiers - IV vasopressor (push dose or drip) → ΔBP to 115/75 - IV NaHCO ₃ bolus → change waveform from wide to narrow QRS complex. Change waveform back to wide QRS after 1 min Triggers - 2 minutes after intubation → 3. PEA
3. PEA Rhythm: Wide QRS complex HR → 122 BP → 0/0 O ₂ SAT → 0% no waveform. EtCO ₂ = 0, no waveform	Unresponsive, GCS = 3	Learner Actions ☐ CPR ☐ Epinephrine ☐ Applies approach to PEA (Rhythm based or Hs & Ts) ☐ Bedside Ultrasound ☐ IV NaHCO ₃ Bolus ☐ If intubated, applies intubation-associated causes of PEA (ex. DOPES mnemonic)	Modifiers - IV NaHCO ₃ bolus → change waveform from wide to narrow QRS complex <u>with a pulse</u> → change rhythm back to wide complex <u>PEA</u> after 1 minute Triggers - Total of 6 amps or 4 amps + infusion of NaHCO ₃ → 4. Resolution
4. Resolution Rhythm: Sinus Tachycardia HR → 111 BP → 109/71 O ₂ SAT → 97% EtCO ₂ = 58		Learner Actions ☐ Repeat EKG ☐ Poison Centre Consultation ☐ ICU Consultation ☐ CT Head ☐ ± Intralipid 20% 1.5ml/kg	END SCENARIO PRN

Tricyclic Antidepressant Overdose



Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results						
Na: 140	K: 3.9	Cl: 100	HCO ₃ : 23	BUN: 4.3	Cr: 40	Glu: 6.5
VBG	pH: 7.29	PCO ₂ : 50	PO ₂ : 45	HCO ₃ : 23	Lactate: 2.1	
WBC: 15.0		Hg: 140		Hct: 0.40		Plt: 220

Tricyclic Antidepressant Overdose



6

Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
Educational Goal:	To recognize and treat and patient presenting after an intentional TCA overdose.
CRM Objectives:	1) Allocate attention wisely between vital signs, history, physical exam, EKG and treatment response in the setting of TCA toxicity. 2) Prevent and manage fixation error in the setting of a patient with altered mental status and fever by maintaining and addressing a wide differential diagnosis.
Medical Objectives:	1) Recognize and appropriately manage a patient in pulseless electrical activity (PEA) arrest. 2) Apply an approach, either rhythm-based or the traditional Hs & Ts, to aid in the management of the underlying cause of PEA. 3) Administer sodium bicarbonate therapy with appropriate treatment targets in the treatment of TCA toxicity with ECG findings of sodium channel blockade.
Sample Questions for Debriefing	
1) What are strategies a team leader can employ to avoid fixation error? 2) What is the differential diagnosis for a patient with altered mental status and fever? 3) Describe a rhythm-based approach to PEA (narrow vs. wide complex). Contrast this approach with the traditional Hs & Ts. 4) What are the EKG and pH targets to guide NaHCO ₃ therapy in TCA toxicity?	
Key Moments	
Supplementary history	
EKG findings of sodium channel blockade	
PEA after intubation	

